Guang-Xing Li, Ph.D.

Senior Scientific ML Researcher | AI for Science & ML Systems

Self-supervised learning · representation learning · PyTorch · GPU/HPC · multiscale physical fields

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Technical Summary

Computational physicist and ML engineer building scientific-ML systems for high-dimensional data, including latent predictive models, from representation architecture and PyTorch implementation through GPU training, dense inference, validation, and open-source release. Creator of ScaleAware-JEPA, a world-model-inspired JEPA system for discovering structure in continuous physical fields.

Technical Skills

- **ML/AI:** Python, PyTorch, self-supervised learning, JEPA, ConvNeXt, representation learning, PCA/UMAP
- **ML Systems:** GPU training, gradient accumulation, checkpointing, tiled inference, config-driven experiments, testing
- **Scientific Computing:** NumPy, SciPy, HPC/Linux, large-scale simulation, high-dimensional scientific data
- **Software:** Git, open-source packaging, reproducible pipelines, documentation

Flagship ML System

ScaleAware-JEPA — *Latent predictive model for scientific discovery* **End-to-end developer** · 2026 · Sole author

Paper: ScaleAware-JEPA: Latent Representation for Discovery in Multiscale Physical Fields · arXiv:2606.29723

Code: github.com/gxli/SA-JEPA

- **Problem:** discover meaningful structure in complex physical data **without predefined labels, objects, or segmentation rules**.
- **Architecture:** predicts hidden structure **in latent space rather than reconstructing raw data**; CDD provides pixel-registered scale components and physical-scale coordinates, while a ConvNeXt V2-style encoder fuses information across scales and produces **dense 32-D latent fields**. Scale-aware masking adapts the prediction context to the intrinsic scale hierarchy rather than fixed image patches.
- **Engineering:** config-driven PyTorch training with gradient accumulation, checkpointing, tiled inference, PCA/UMAP diagnostics, tests, scripts, documentation, and reproducible experiment configs.
- **Validation:** evaluated on a $256 \times 256 \times 256$ **MHD turbulence cube**, molecular-gas observations, and urban nighttime-light data; learned dense structural atlases recover coherent morphology and **physically distinct scaling relations** without predefined labels or segmentation.

Technical Ownership & Collaboration

- **Own SA-JEPA end to end:** architecture, PyTorch implementation, experiment design, validation, testing, documentation, release.
- Maintain open-source scientific-ML codebases with reproducible configs, tests, examples, and documentation.

Scale-Aware Adversarial Analysis — *Generative-model diagnostics* 2026 · Co-author

Paper: Scale-Aware Adversarial Analysis: A Diagnostic for Generative AI in Multiscale Complex Systems · arXiv:2605.00510

- **Method:** deterministic, physically constrained interventions in CDD scale space for evaluating DDPMs of multiscale systems.
- **Result:** exposed localized structural freezing and nonlinear instability missed by aggregate image-quality metrics.

Additional Technical Projects

Adjacent Correlation Analysis (ACA) — *Discovery of hidden regularities* 2025 · Sole author

Paper: Revealing Hidden Correlations from Complex Spatial Distributions: Adjacent Correlation Analysis · arXiv:2506.05759

Code: github.com/gxli/Adjacent-Correlation-Analysis

- **Method:** recovers local relationships hidden by global statistics, representing complex spatial patterns as phase-space vector fields.
- **Result:** recovered vectors form the vector field on an attracting manifold— enabling classification, prediction, parameter fitting, and forecasting.

Constrained Diffusion Decomposition (CDD) — *Multiscale decomposition for continuous physical data* 2022 · Sole author

Paper: Multi-scale Decomposition of Astronomical Maps: A Constrained Diffusion Method · ApJS 259, 59 · arXiv:2201.05484

Code: github.com/gxli/Constrained-Diffusion-Decomposition · PyPI: constrained-diffusion

- **Method:** nonlinear-diffusion decomposition producing pixel-registered physical scales without the ringing and negative artifacts introduced by conventional band-limited filtering.
- **Reuse:** multiscale representation used as an input coordinate for SA-JEPA and scale-aware generative-model diagnostics.

Experience

Principal Investigator & Associate Professor **2018–Present**
Yunnan University (SWIFAR), China

- Lead computational research projects spanning scientific ML, multiscale modeling, and high-dimensional observational and simulation data.
- Supervise PhD/MSc researchers in computational methods, scientific data analysis, and research software development.
- Advise collaborators on complex observational and simulation datasets, translating scientific questions into computational formulations, physical diagnostics, data pipelines, and ML workflows.

DFG Research Fellow **2014–2018**
LMU Munich (Universitäts-Sternwarte), Germany

- Led a DFG-funded program developing computational methods to extract physical understanding from observational and simulation data in Linux/HPC environments.
- Developed G-virial, combining analytical physics with numerical simulations for structure identification in turbulent data.

Ph.D. Researcher **2010–2014**
Max Planck Institute for Radio Astronomy, Germany

- Built scalable algorithms for multiscale structure extraction from high-dimensional, heterogeneous datasets.

Research Impact

111 scientific publications · 1,800+ citations · h-index 24 · €280k+ PI-led competitive funding

Education

Ph.D. Astrophysics / Computational Modeling — Max Planck Institute for Radio Astronomy & University of Bonn (2014), *magna cum laude*
M.Sc. Astrophysics — University of Science and Technology of China (2010)
B.Sc. Electronics — Peking University (2007)